

Lab Assignment 10: Clustering Medical Images and Crop Disease Classification

CSAI2017P: Applied Machine Learning (Lab)

Experiments 13–14 (Units VI–VII) • CO2, CO4

Instructor: Dr. Tofik Ali**Due:** announced in the lab**Student Name:** _____**SAP ID:** _____**Batch:** _____**Lab quiz:** LQ10, after submission

Objective

- Cluster image data without labels and evaluate the result honestly.
- Choose the number of clusters with a defensible criterion.
- Build an image classifier and diagnose it from its errors.

Datasets used in this assignment

B1 (digits), B3 (Breast Cancer), B4 (PlantVillage leaf subset).Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA10_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)**A1. Cluster the digits** [2 marks]

Dataset: B1 (digits), labels withheld.

- (a) Scale the features and run K-means with $k = 10$.
- (b) Report inertia and the silhouette score.
- (c) State in two lines why you cannot report accuracy here.

A2. Choosing k [2 marks]

Dataset: B1 (digits).

- (a) Run K-means for $k \in \{2, \dots, 15\}$ and plot inertia and silhouette against k .
- (b) State the k each criterion suggests.
- (c) In two lines, say what you would conclude if the two criteria disagreed.

A3. Load the leaf images [2 marks]

Dataset: B4 (PlantVillage tomato subset).

- (a) Load the subset, resize to 64×64 and report the number of images per class.
- (b) Display one image per class in a grid.
- (c) State in two lines whether the classes are balanced and what you would do if not.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)**B1. Do the clusters mean anything** [4 marks]

Dataset: B1 or B3, your choice.

- (a) Cluster the data at your chosen k , then compare the clustering with the withheld labels using the adjusted Rand index and a cluster-versus-label crosstab.
- (b) Report ARI and identify one cluster that is nearly pure and one that mixes classes.
- (c) In 6–8 lines, explain what ARI measures and why a low ARI does not automatically mean the clustering is useless.

B2. Leaf disease classifier [4 marks]

Dataset: B4.

- (a) Build a classifier on flattened resized images or on simple colour-histogram features; use a stratified split.
- (b) Report macro-F1 and the per-class F1 table.
- (c) Show four misclassified leaves and explain in 6–8 lines what confuses the model.

Section C (Hard) — compulsory (2)**C1. What would you tell the farmer** [2 marks]

Dataset: B4.

- (a) State your model's macro-F1 and its worst per-class F1.
- (b) In 4–6 lines, write what you would honestly tell someone about to rely on this model in a field, including one thing it will get wrong.

End of Lab Assignment 10

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.